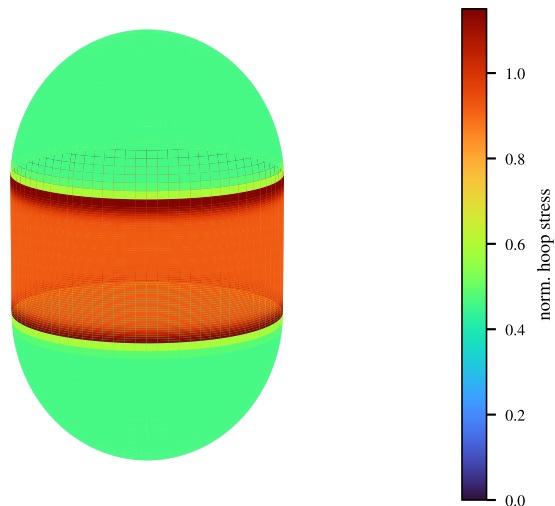
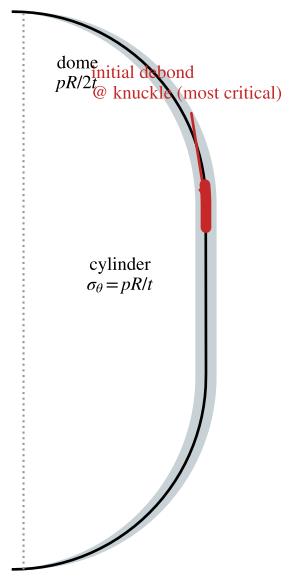


CFRP COPV tank (5th structure, internal-pressure mode): 3-D vessel, layup section, real CZM-FEM p_{cr}

(a) COPV pressure vessel: hoop-stress field
(cyl pR/t , dome $pR/2t$, knuckle conc.)
 $\pm 54.7^\circ$ winding, initial debond (red)



(b) meridional section: cyl $\rightarrow$ knuckle $\rightarrow$ dome,
mid-wall interlaminar interface + debond



(c) real CZM-FEM: p_{cr} vs damage
(junction most critical; dome strongest)

